# Employee management system

A PROJECT REPORT ON

Employee Management System

**A PARTIAL FULFILLMENT OF:**

BCA SEM-6

**SUBMITTED TO:**

Geetanjali College of Computer Science and Commerce

**AFFILIATED TO:**

SAURASTRA UNIVERSITY

**ACADEMIC YEAR:**

2025-2026

**UNDER THE GUIDANCE OF:**

PROF. Jigisha Ganatra

**PREPARED BY:**

Davera Hardik

# ACKNOWLEDGEMENT

While the report for my project work is ready, I feel it is my duty to offer my sincere thanks to all those who have helped and guided me in this regard.

Action is very necessary to convert our dreams into reality. My dream was to develop the project. To make a successful and useful project, one needs help, understanding, co-ordination from all those who are directly or indirectly involved in this.

Many people have contributed to make this project a reality. I would like to express my gratitude to our **Head of Department "Prof. Brijesh Shah"** for this guidance throughout the project.

While submitting the project report it is my duty to offer my sincere thanks to those who have assisted me in preparing this project report work.

Before I describe in brief about my project, I would like to add a few heartfelt words for those people who had given their time and knowledge to developing this project.

I am thankful to all my college faculties and friends for their kind co-operation.

Your Faith Fully,
Davera Hardik

# PREFACE

In the 21st Century of Technology, Computers and related services are the most useful, automatic and highly speed of work. B.C.A is a course in which the practical aspect is as important as the theoretical aspect. In the present world, the importance of education is increasing. Project training in a student's life is like a live experience in the industry. If we have experience then & then we get top of the level.

The most important factor of a B.C.A Course is Project Work. The Project Work is Must for Every Student in a Different Government or Private Organization during the Study Period.

I am developing a project in the 6th Semester of B.C.A, I am a student of Geetanjali Group of Colleges affiliated with Saurashtra University, Rajkot.
I am creating this project dynamic and, in this project, we provide insert and delete operation on admin side and select at client side so the user can easily show the site.

I have gone through and studied necessary information in formation, which is collected and presented in this report. I am trying my best in this project. I have taken almost care that the information provided in the project is true and the project is in a smooth-running condition and error free.

# PROJECT PROFILE

- Project Title :- Employee Management System
- Development Software :- Visual Studio 2022
- Front End :- Python
- Back End :- MySQL
- Academic Year :- 2025-2026
- Developed By :- Davera Hardik
- Submitted To :- Geetanjali Group of College
- Documentation Tool :- Microsoft Word
- Operating System :- Window 11

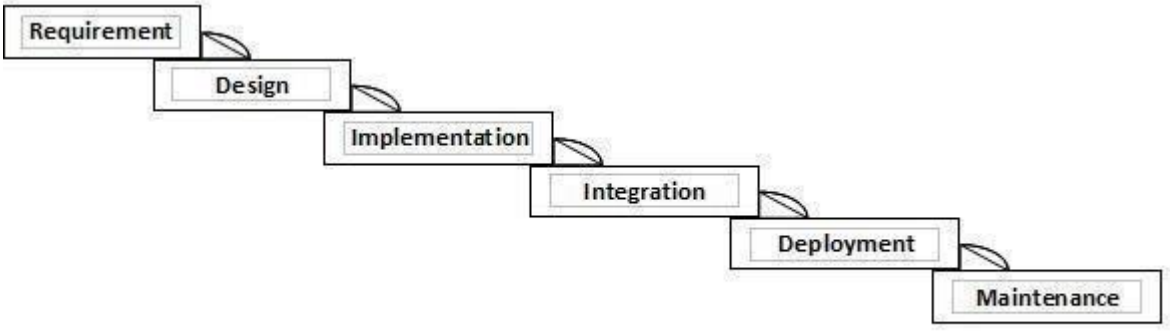
# PROCESS MODEL

The Process Model used in our project "College Management System" is a waterfall model.

## ➢ The Waterfall Model:

The waterfall model is a sequential design process, used in software development processes, in which progress is seen as flowing steadily downwards (like a waterfall) through the phases of Conception, Initiation, Analysis, Design, Construction, Testing, Production/Implementation and Maintenance.

The waterfall development model originates in the manufacturing and construction industries: highly structured physical environments in which after-the-fact changes are prohibitively costly, if not impossible. Since no formal software development methodologies existed at the time, this hardware-oriented model was simply adapted for software development




Waterfall model

## ➢ Reason the waterfall model in the software development cycle:

- ➔ Since we have well known, clean and fixed requirements therefore it best suits our software development.
- ➔ Our product definitions are stable.
- ➔ Technology is clearly understood.
- ➔ The project is short.

## ➢ Advantages of the waterfall model:

- ➔ This model is simple and easy to understand and use. In this model phases are processed and completed one at a time and phases do not overlap.

- ➔ Waterfall model works well for smaller projects where sequence is very well understood

# PROJECT INFORMATION

In the daily Employee Management System, we can use Python, MYSQL server Database and many packages. This is the project which can be used for Manage employee's role vice and handle their issues.

- **Admin Module:**
  1. **Register Employee:**
     In this section, Admin can only register new employees (add).
  2. **Employee:**
     In this section admin show employees details which were added by them and manage employees (edit, delete)
  3. **Leave:**
     In this section admin can approve leave and edit leave applications which employees registered by them and they can edit and delete their application.
  4. **Salary:**
     In this section admin can assign salaries to employees and also resolve corrections in salaries of employees.
  5. **Logout:**
     Through this section admin logout from the website

- **Employee:**
  1. **Profile:**
     In this section employees can view their details
  2. **Leave:**
     In this section can view their applied leaves status and edit their applied leaves if not have been accepted or rejected by superadmin and admins
  3. **Salary:**
     In this section employees can download their salary slip and also request for corrections if have mistaken in their salary.
  4. **Logout:**
     Through this section employees logout from system

# SYSTEM REQUIREMENTS

## Hardware Requirements:

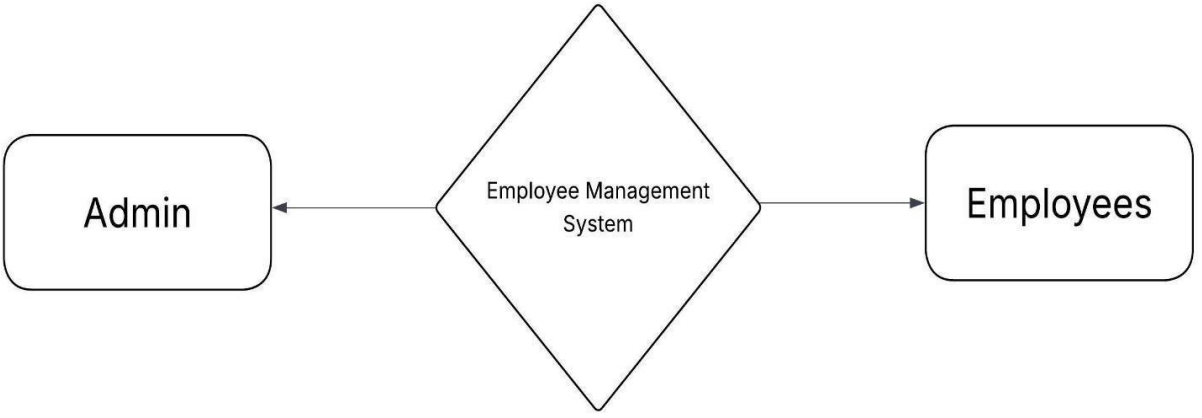
| | |
|---|---|
| Processor | Intel(R) Core (TM) i5-835OU CPU @1.70Ghz 1.90Ghz |
| Ram | 16.0GB (15.9GB) usable |
| Hard disk | 64-bit operating system x64-based processor |
| Operating system | Windows-11 |

## Software Requirements:

| | |
|---|---|
| Front End tool | Python |
| Back-end tool | MySQL |
| Editing tool | Visual Studio code |
| Browser | Internet Explorer or Mozilla Firefox or Google Chrome |

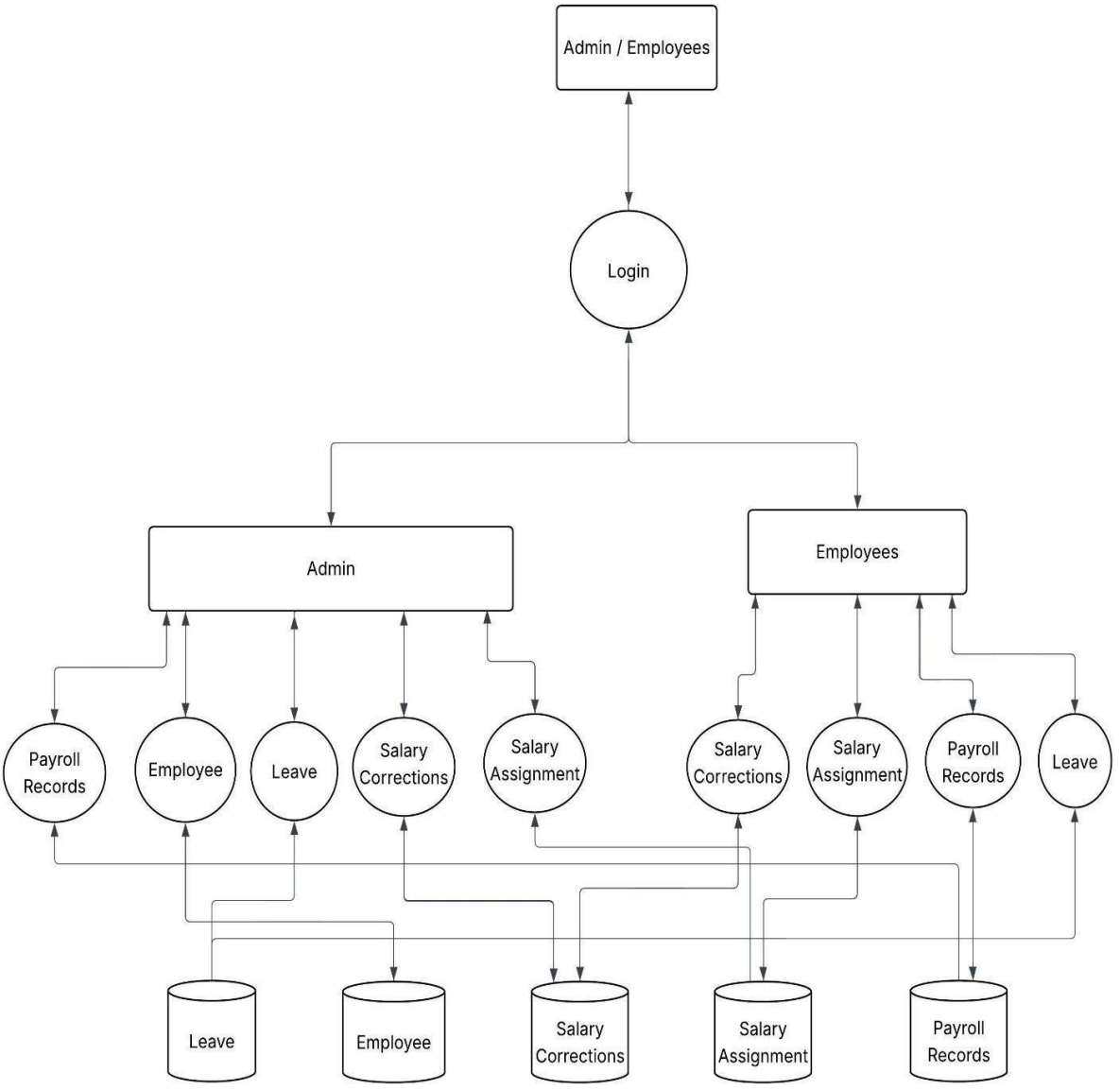
# DATA FLOW DIAGRAM

## ➢ CONTEXT LEVEL DFD: -

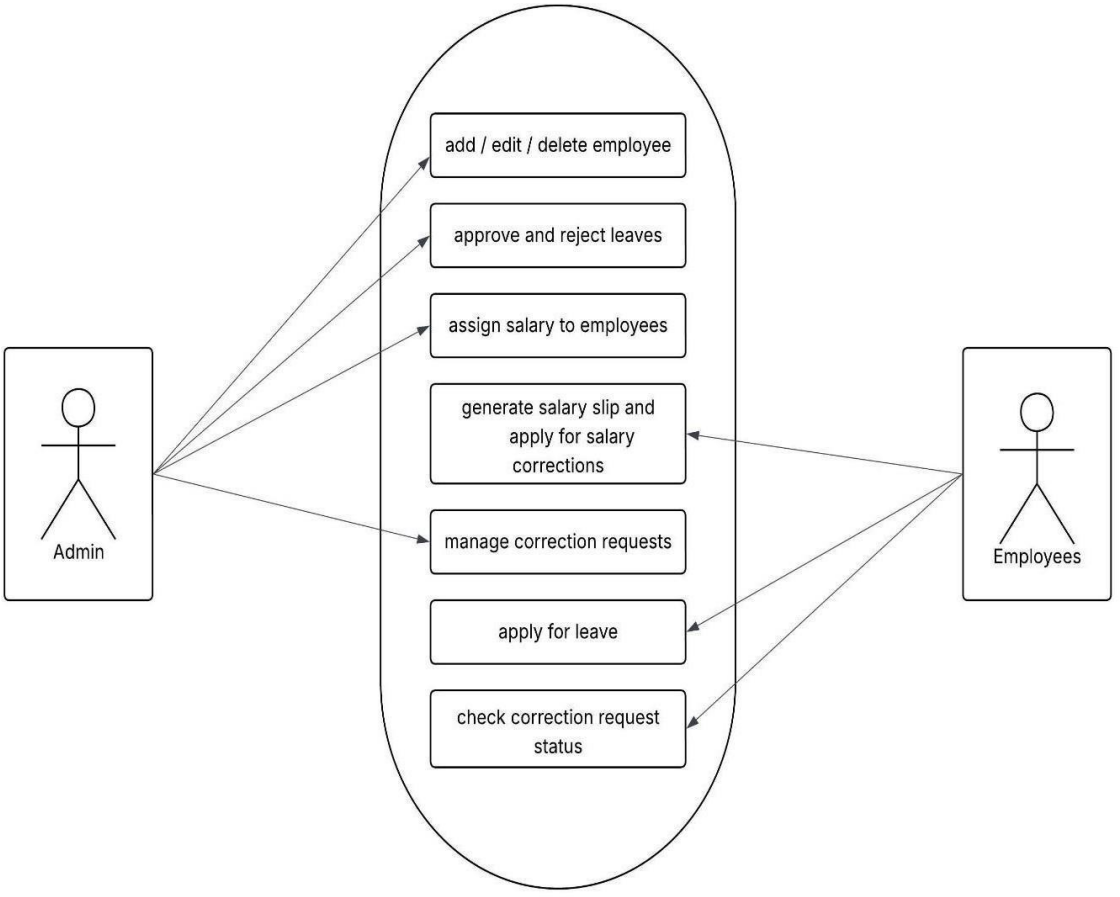


# Employee management system

## ➢ ONE LEVEL DFD:



# USE CASE DIAGRAM



# DATA DICTIONARY

A data dictionary is a set of information describing the contents, format and structure of a database and the relationship between its elements, used to control access to and manipulation of the database.

- **DATABASE TABLES:**
  - employees
  - leaves
  - payroll_records
  - salary_assignments
  - salary_corrections

## 1. Employees:-

| Field name | Data type | Size | Constraint key |
|---|---|---|---|
| Id | Varchar | 10 | PRIMARY KEY |
| Name | Varchar | 50 | Not NULL |
| Password | Varchar | 50 | Not NULL |
| Phone | Varchar | 10 | Not NULL |
| Email | Varchar | 200 | Not NULL |
| Salary | double | 20 | Not NULL |
| DOB | Date | - | Not NULL |
| Department | Varchar | 10 | Not NULL |
| Role | Varchar | 10 | Not Null |

# Employee management system

## 2. Leaves:-

| Field name | Data type | Size | Constraint key |
|---|---|---|---|
| Id | Varchar | 10 | PRIMARY KEY |
| emp_id | Varchar | 50 | Not NULL |
| emp_name | Varchar | 50 | Not NULL |
| Leave_type | Bigint | 10 | Not NULL |
| Start_date | date | - | Not NULL |
| End_date | Date | - | Not NULL |
| Reason | Text | - | Not NULL |
| Status | Varchar | 32 | Not NULL |
| Applied_date | Datetime | - | Not NULL |
| Duration_type | Varchar | 32 | Not Null |

## 3. Salary_assignments:-

| Fieldname | Data type | Size | Constraint key |
|---|---|---|---|
| Id | Varchar | 64 | PRIMARY KEY |
| Emp_id | Varchar | 64 | Not NULL |
| Assigned_salary | Double | - | Not NULL |
| Assigned_on | Datetime | - | Not NULL |
| Assigned_by | Varchar | 64 | Not NULL |
| Month | Varchar | 7 | Not NULL |

## 4. Payroll_records:-

| Fieldname | Data type | Size | Constraint key |
|---|---|---|---|
| Id | Varchar | 64 | PRIMARY KEY |
| Emp_id | Varchar | 64 | Not Null |
| Month | Varchar | 7 | Not Null |
| Base_salary | Double | - | Not Null |
| Overtime_hours | Double | - | Not NULL |
| Bonus | Double | - | Not NULL |
| Other_deductions | Double | - | Not NULL |
| Leave_deduction | Double | - | Not NULL |
| Net_salary | Double | - | Not NULL |
| Generated_on | Datetime | - | Not NULL |
| Slip_path | Varchar | 255 | Not NULL |

## 5. Salary_corrections:-

| Fieldname | Data type | Size | Constraint key |
|---|---|---|---|
| Id | Varchar | 64 | PRIMARY KEY |
| Emp_id | Varchar | 64 | Not NULL |
| Month | Varchar | 7 | Not NULL |
| Description | Text | - | Not NULL |
| Submitted_on | Datetime | - | Not NULL |
| Status | Varchar | 32 | Not NULL |
| Assignment_id | Varchar | 64 | Not NULL |
| Payroll_id | Varchar | 64 | Not NULL |

# TEST CASES

The testing phase is an important part of the software developed. It is the process of finding errors and missing operations and a complete verification to determine whether the objectives are met and the user requirements are satisfied.

## ➢ Login Screen:

| TestCase ID | Module | Test Scenario | Test Steps | Input Data | Expected Result | Actual Result | Status |
|---|---|---|---|---|---|---|---|
| TC001 | Login | Valid Login | Enter valid admin id& password | Adminid: EMP001 Password: admin@123 | User redirectedto Home page | As expected, | Pass |
| TC002 | Login | Invalid login | Enter valid id & wrong password | Admin id: EMP001, Password: xyz | Errormessage shown | As expected, | Pass |
| TC003 | Login | Valid Login | Enter valid id & password | User id: EMP002 Password: user@123 | User redirect to Loginpage | As expected, | Pass |
| TC004 | Login | Invalid Login | Enter valid id & password | User id: EMP002 Password:xyz | Errormessage shown | As expected, | Pass |

# Employee management system

## ➢ Employee Register Screen for Admins:

| Test Case ID | Module | Test Scenario | Test Steps | Input Data | Expected Result | Status |
|---|---|---|---|---|---|---|
| TC001 | Register | Successful Registration | Fill all fields and click "Register" | Name, Phone, Email, Password ,Date of Birth, Departme nt, role, salary | Account created & confirmation message shown | Pass |
| TC002 | Register | Missing Field Validations | Leave "Email" and 'phone'', ''Password'' empty and click "Register" | Name, Date of Birth, Departme nt, role, salary | Error message: "Fields are required" | Pass |
| TC003 | Register | Duplicate Email and contact no | Try to register with an already registered email and contact number | Email: test@test .com phone: 12345678 90 | Error message: "Email and phone already exists" | Pass |
| TC004 | Register | Invalid Joining Date and Birth of Date | Date of Birth is same, If Birth of date is invalid means age is under 18 | Date of Birth: 28-9-16 | Error Message : ''Enter Valid dates" | Pass |

# Employee management system

## ➢ Employees Listing for Admins:

| Test Case ID | Module | Test Scenario | Test Steps | Expected Result | Status |
|---|---|---|---|---|---|
| TC001 | Employee list | Employees List | Admins: employees listed here which are registered by them. | List of employees and admins shown basis on role | Pass |

## ➢ Show Leave Applications Screen:

| Test Case ID | Module | Test Scenario | Test Steps | Expected Result | Status |
|---|---|---|---|---|---|
| TC001 | Leave (Admin) | Leave Applications List | Admins: Which employees leave applications listed here which are registered by them. | Leave Applications list which applied by employees | Pass |

# Employee management system

## ➢ Employees Edit Screen for Admins:

| Test Case ID | Module | Test Scenario | Test Steps | Input Data | Expected Result | Status |
|---|---|---|---|---|---|---|
| TC001 | Edit | Successful Edit | Fill all fields and click "Edit" | Name, phone, Email, Password, Date of Birth, Departme nt, role, salary | Account Edit & confirmation message shown | Pass |
| TC002 | Edit | Missing Field Validations | Leave "Email" and 'phone'', ''Password'' empty and click "Edit" | Name, Date of Birth, Departme nt, role, salary | Error message: "Fields are required" | Pass |
| TC003 | Edit | Duplicate Email and contact no | Try to Edit with an already registered email and phone | Email: test@test.com phone: 1234567890 | Error message: "Email and phone already exists" | Pass |
| TC004 | Edit | Invalid Birth of Date | If Birth of date is invalid meansage is under 18 | Date of Birth: 28-9-16 | Error Message: ''Enter Valid dates" | Pass |

# Employee management system

## ➢ Leave Screen:

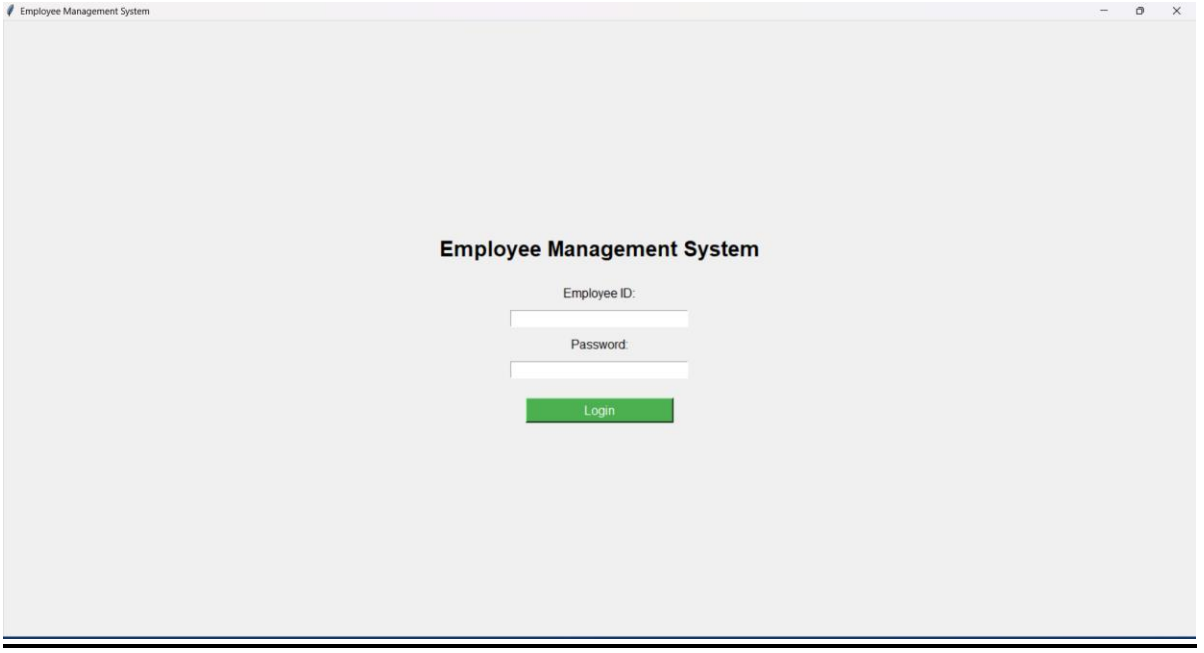
| Test Case ID | Module | Test Scenario | Test Steps | Input Data | Expected Result | Status |
|---|---|---|---|---|---|---|
| TC001 | Leave | Leave Request | Fill Leave Form and click“Apply Leave” | Leave type, leave reason, from date and to date | Leave request submitted successfully and send message | Pass |
| TC002 | Leave | Missing leave type, From and to date | Submit without leave type and from and to date | Leave reason | Error Message: Fill required fields please | Pass |
| TC003 | Leave | Invalid Date range | From date greaterthan to date | From: 29-9-25 To: 28-9-25 | Error Message: Enter Valid dates | Pass |
| TC004 | Leave | Change status | Leave must be approved or rejected | Leavestatus will be accepted or rejected | Success Message: Leave Status change successfully | Pass |

# Employee management system

| | | | | | | |
|---|---|---|---|---|---|---|
| TC005 | Leave | Edit Leave | Leave will not be accepted or rejected | Leave type, Leave Reason, From date and to date | Success Message: Leave Updated successfully | Pass |
| TC006 | Leave | Edit & Delete Leave | Leavestatus will be pending | Edit: Leave Type, Leave reason, from date, To date | Success Message: Edit -> Leave Application Updated successfully, Delete -> Leave application deleted successfully | Pass |

# SNAPSHOTS

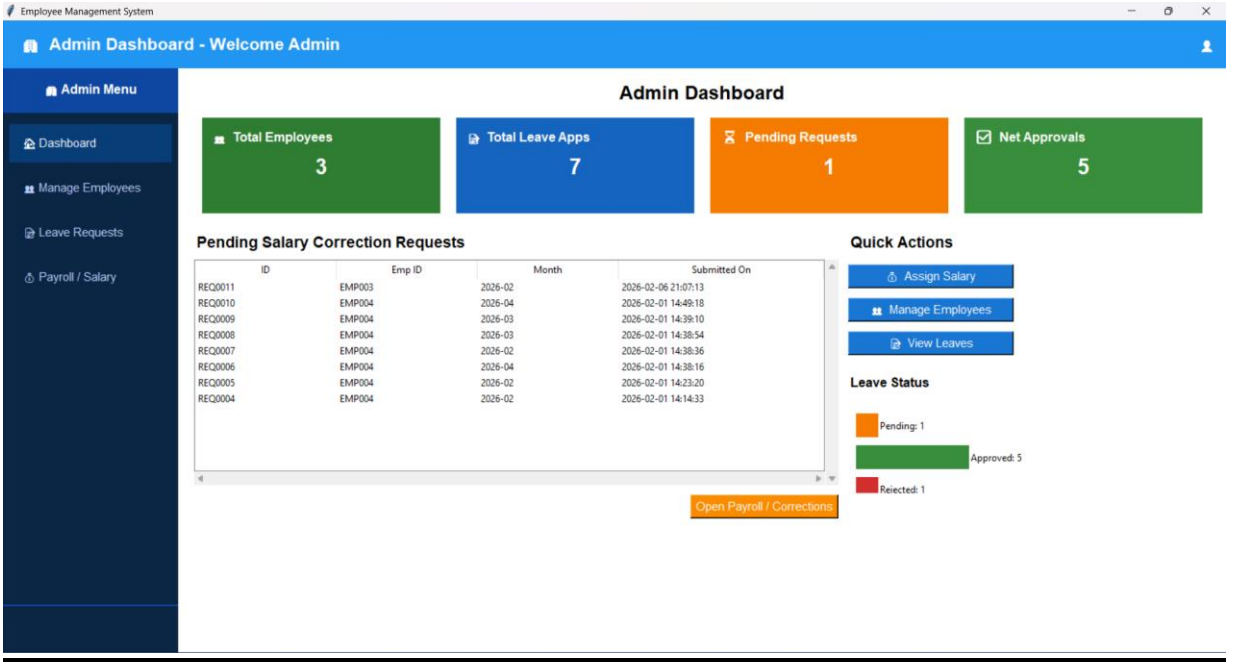
- Employee & Admin login screen




- Employees and admin can login from here and redirect to dashboard.

# Employee management system

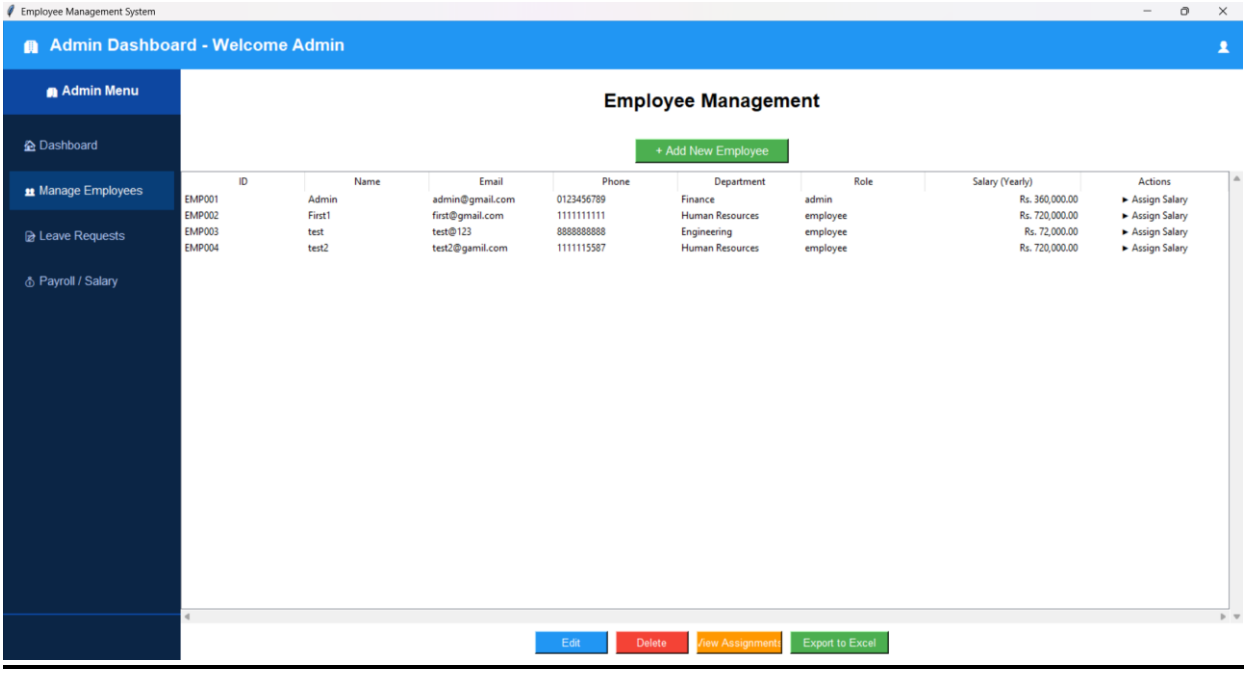
## ➢ Admin Dashboard




- After login admins redirect to their dashboard where show records about employee count and total leaves count with status
- Some quick actions to access that menu from this quick action menu.

# Employee management system

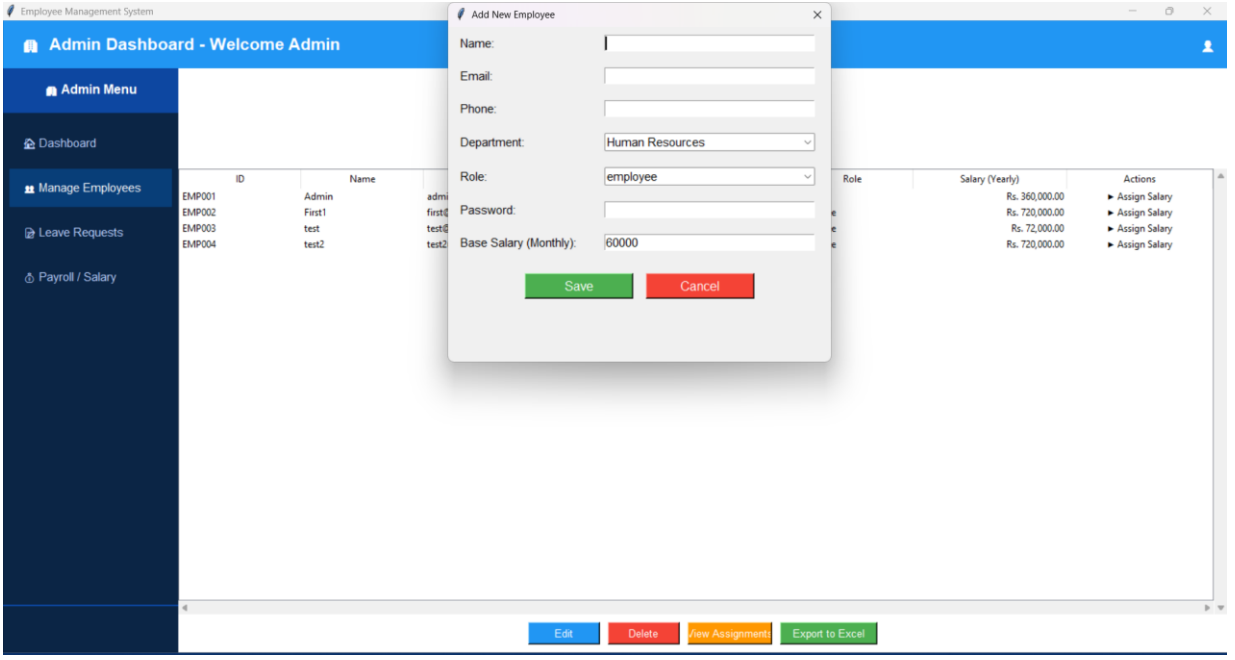
## ➢ Manage Employee Screen




- Here admin can perform CRUD for employee
- Admin can also assign salary to employees and view assigned salary of employees
- Also export this employee's data into excel.

# Employee management system

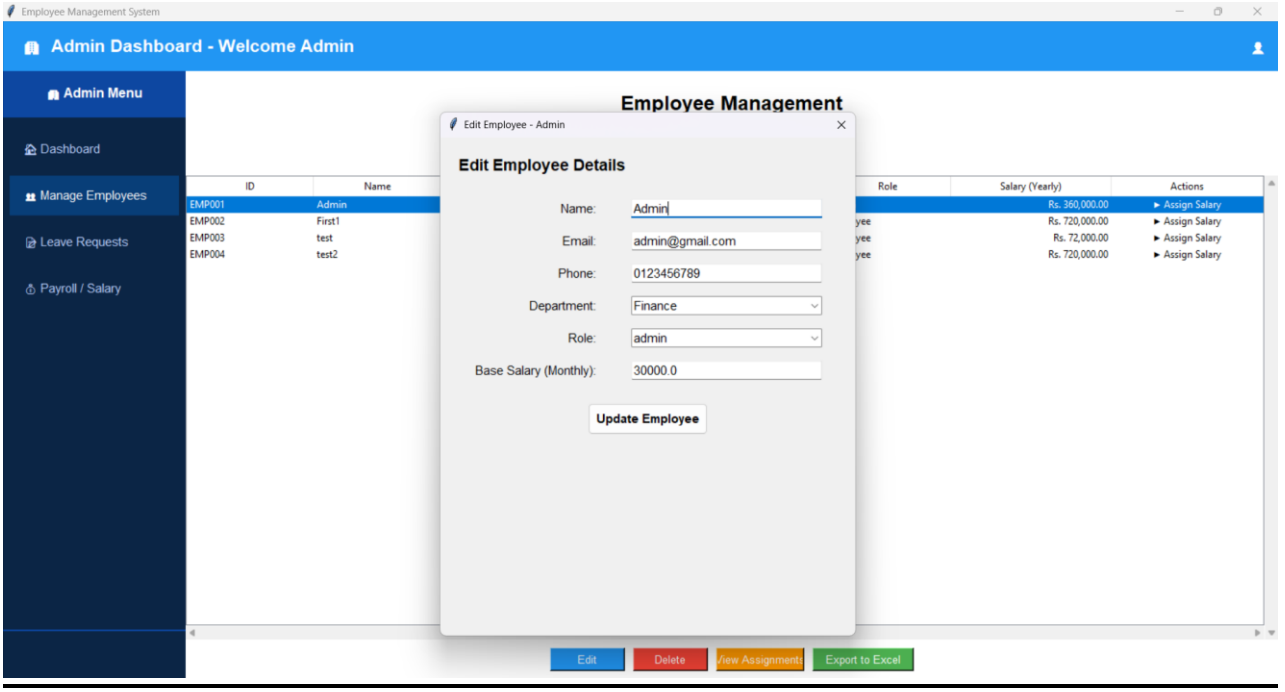
## ➢ Create employee module




- Admin can add new employee using this modal

# Employee management system

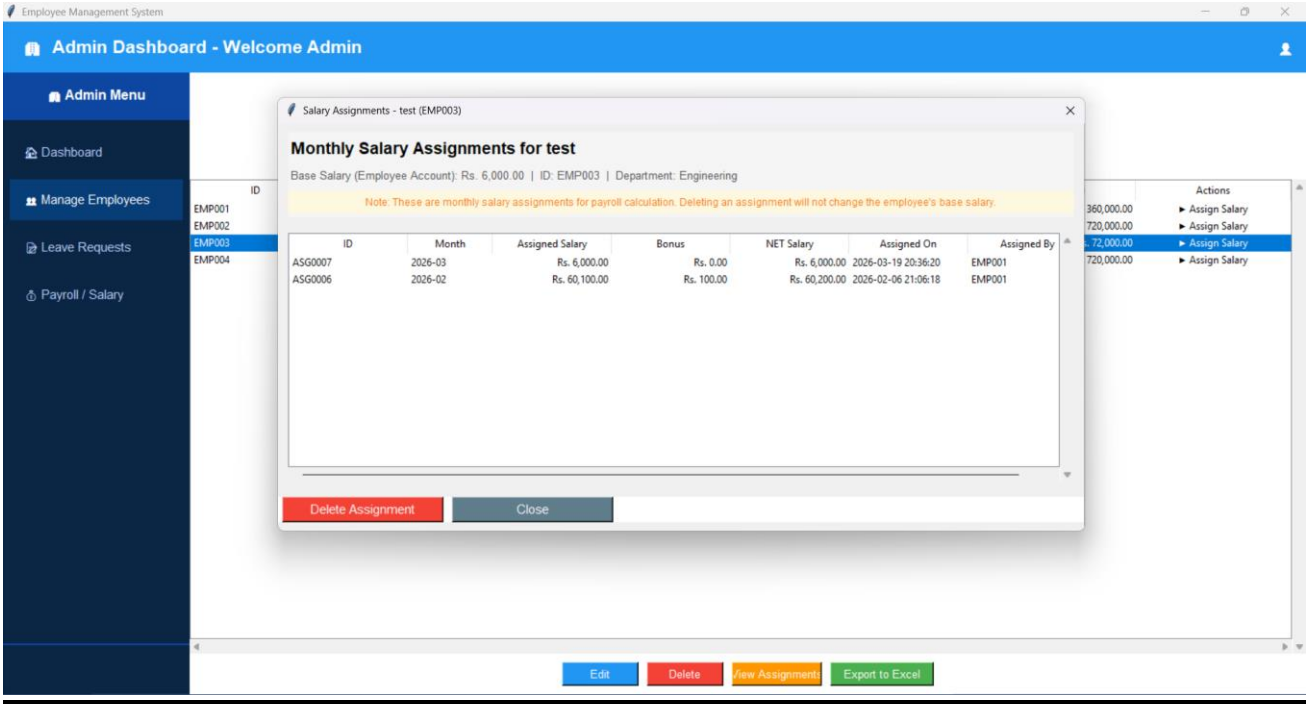
## ➢ Edit employee module




- Admin can update employee details using this modal.

# Employee management system

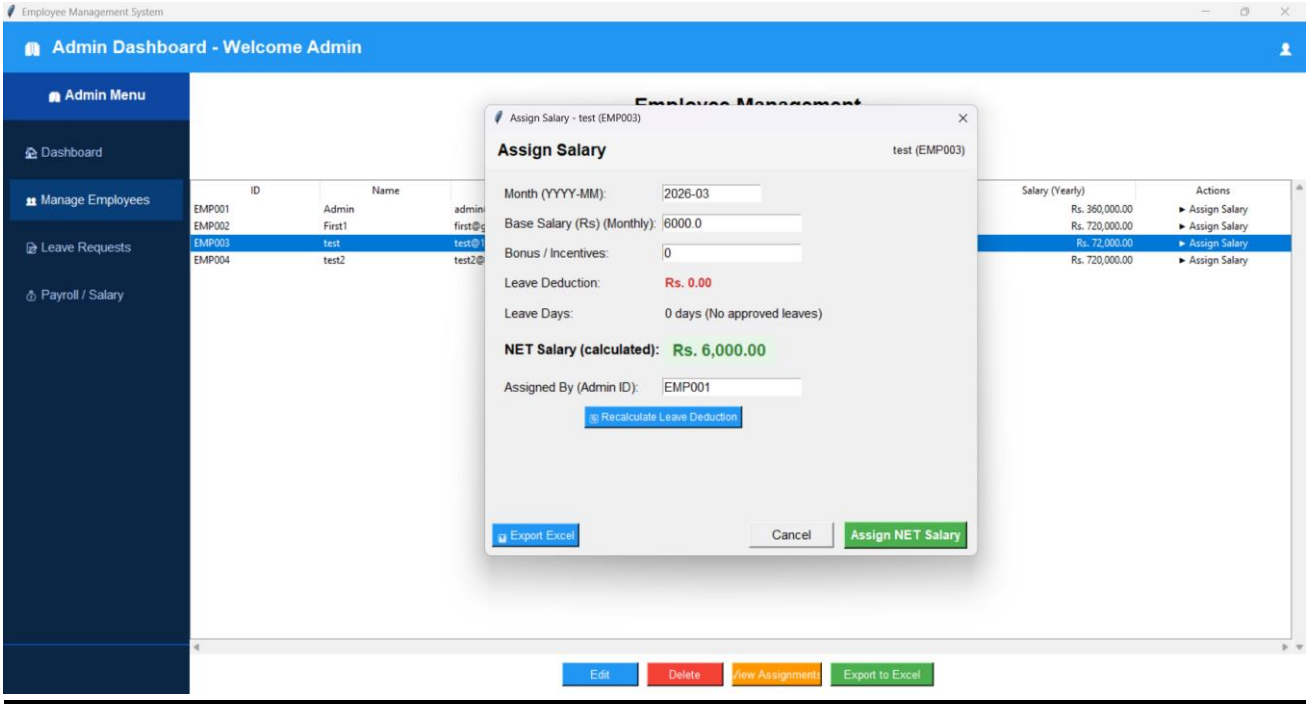
## ➢ Assign salary records module




- Here in this dialog box admin can show the assigned salary to employee
- Also, admin can delete assigned salary to employee if have correction in salary

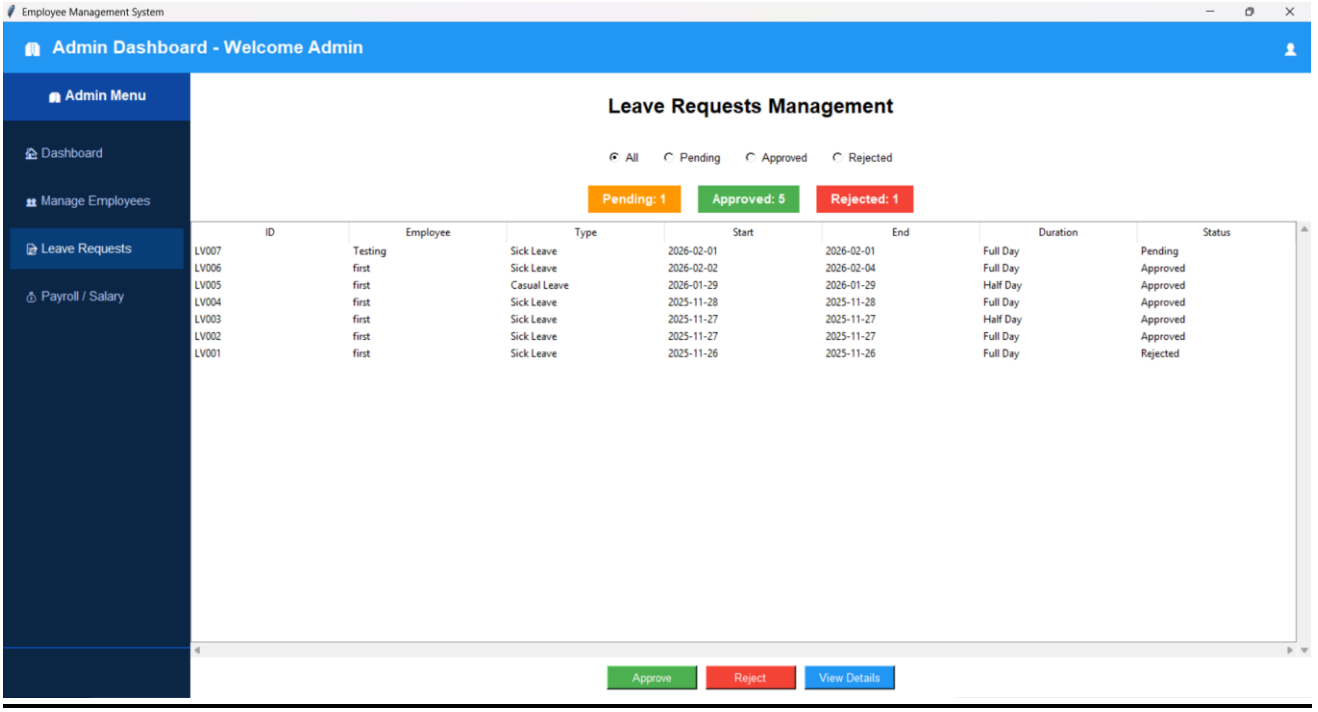
# Employee management system

## ➢ Assign salary module




- Admin can assign salary using this modal and also admin can export the assigned salary

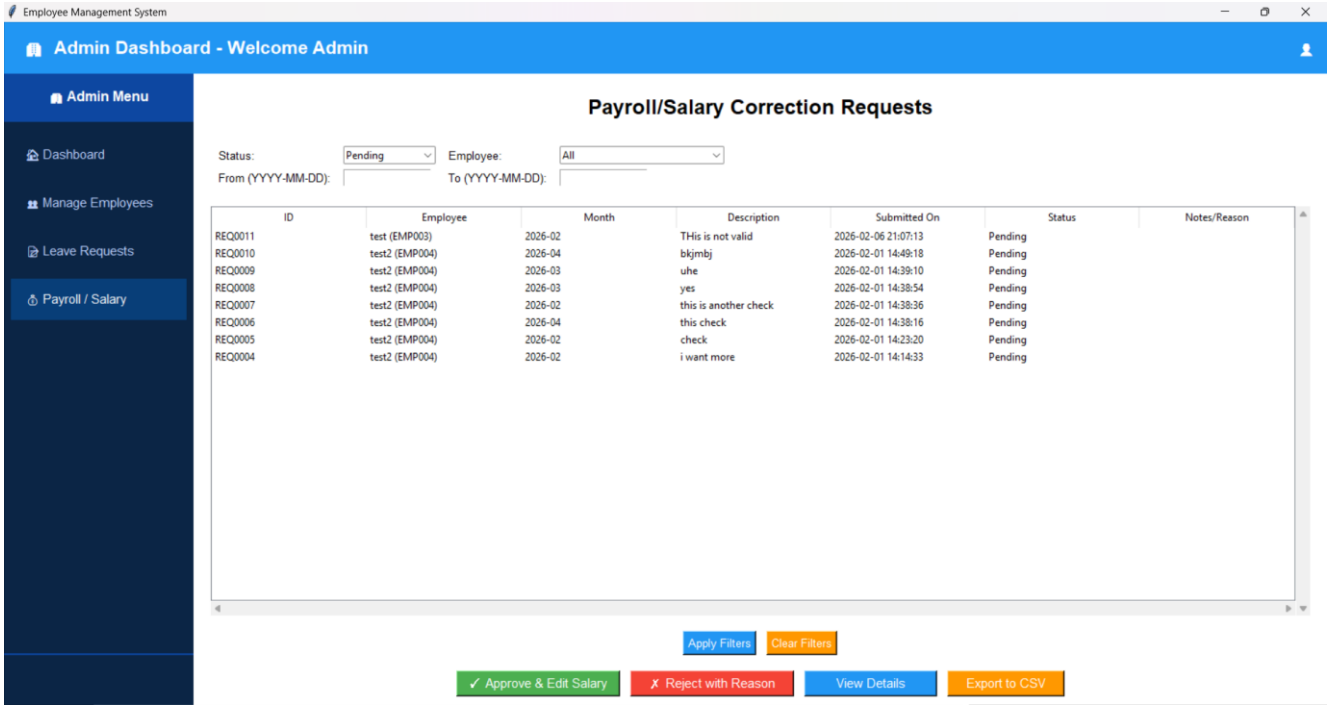
## ➢ Leave Requests Module




- Here admin can show the leave applications of employees
- Also, admin can accept and reject employees leave applications

# Employee management system

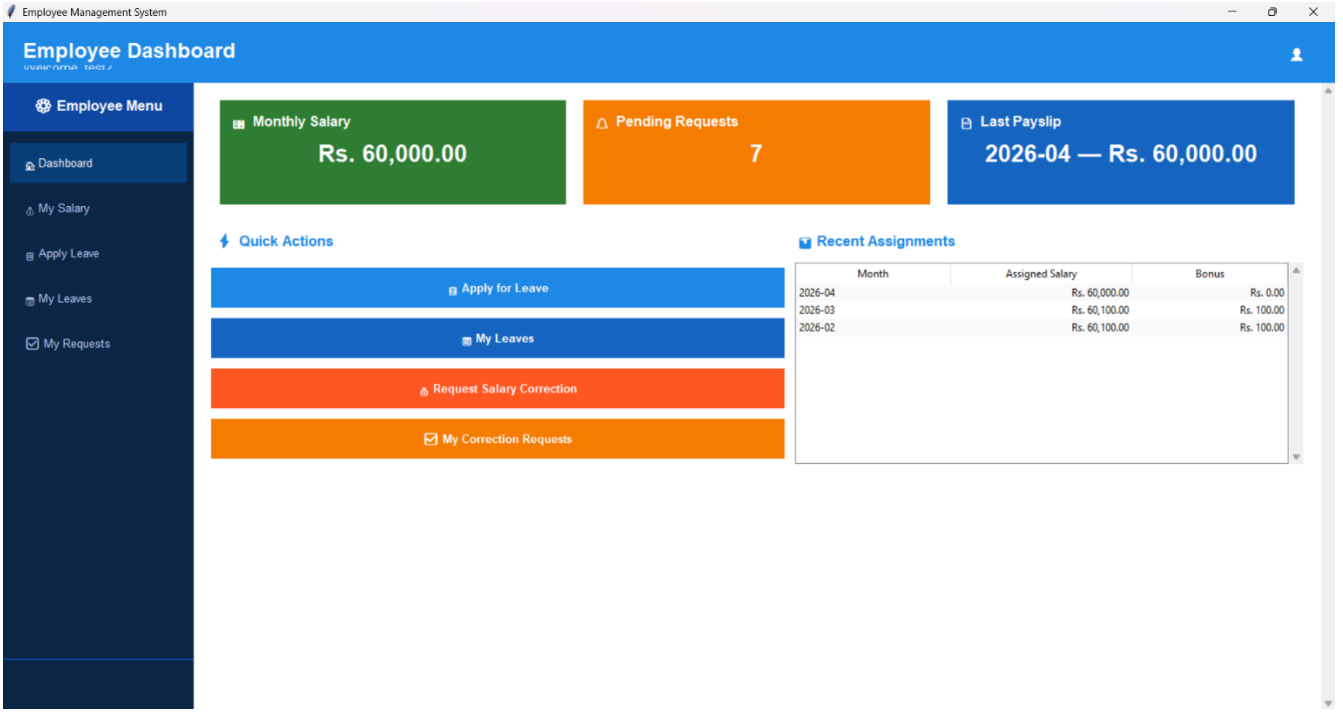
## ➢ Salary correction requests module




- Here admin can see list of salary correction requests of employees
- Also, admin can accept and reject this salary correction request with reasons
- We can also export these details in excel

# Employee management system

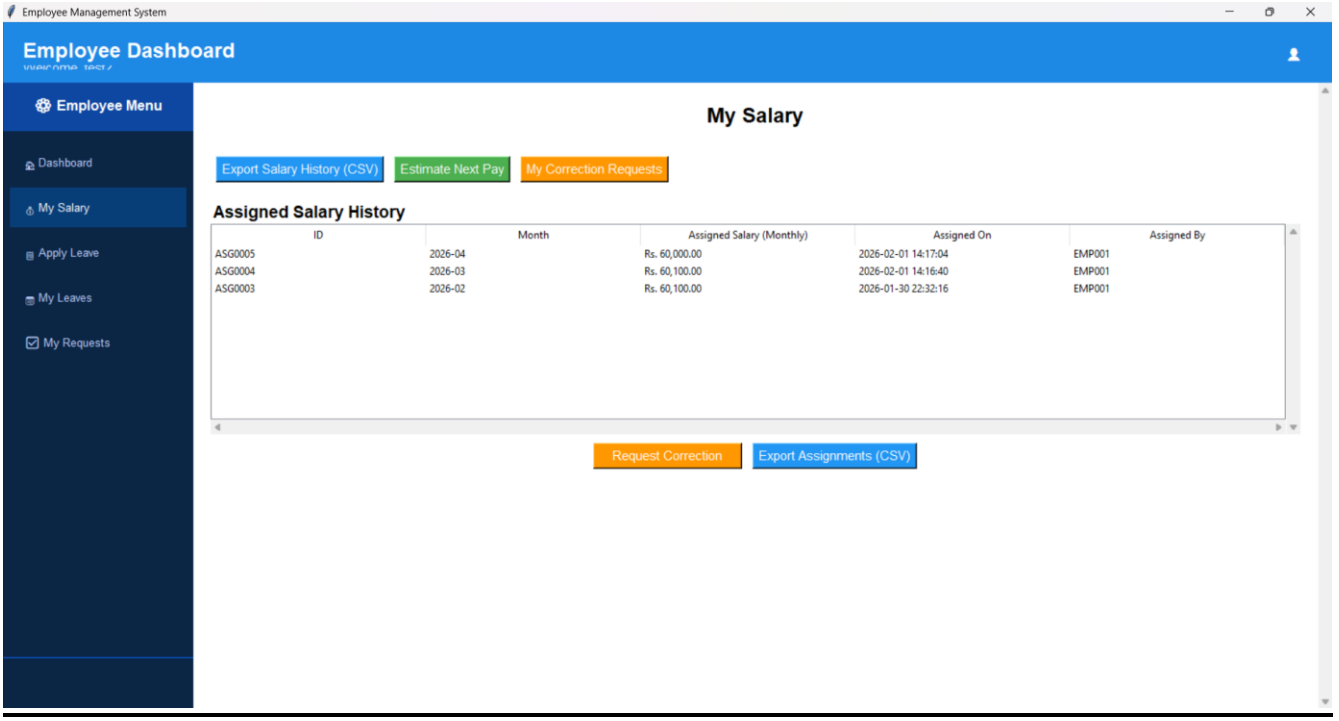
## ➢ Employee dashboard




- After login employees can redirect to this dashboard where they can see their current salary.
- See their recent assigned salary and also quick actions to access some functionality directly.

# Employee management system

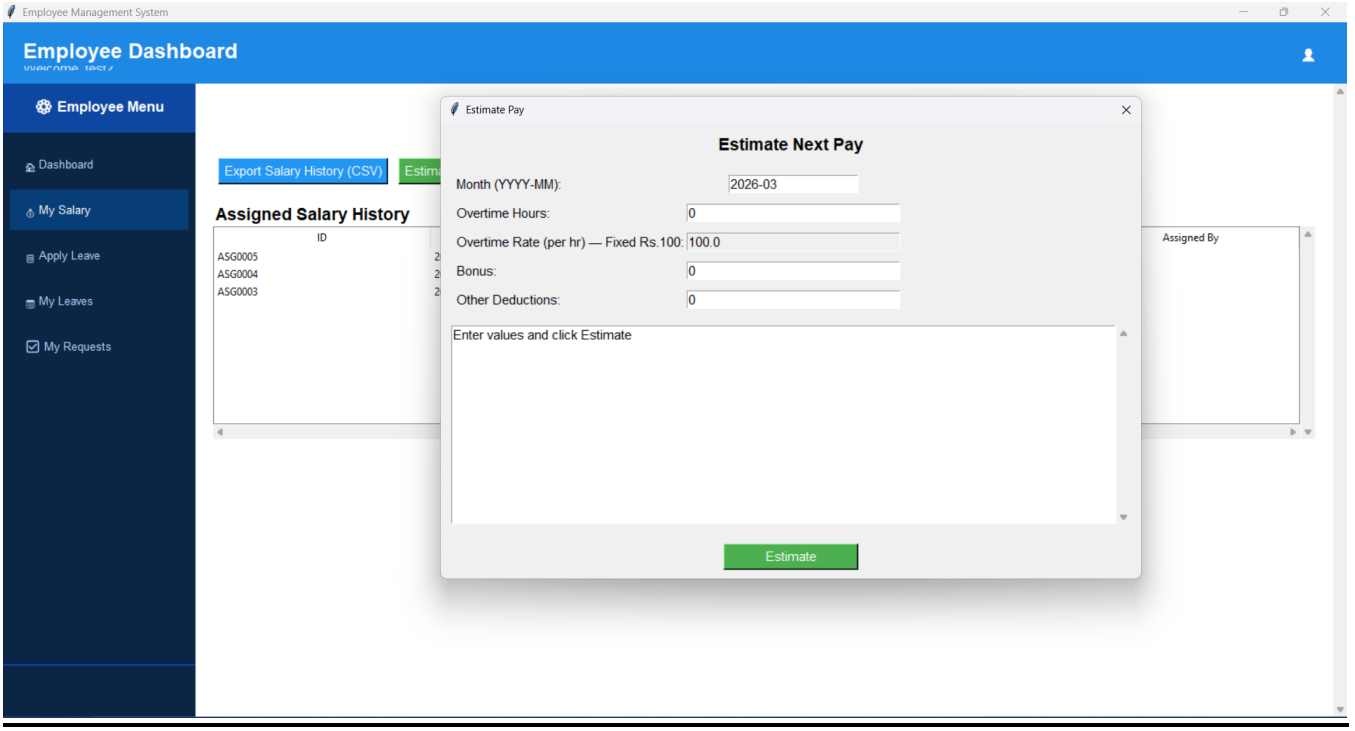
## ➢ Employee salary module




- Here employees can see their assigned salary
- If employees have issue in their assigned salary so they request for correction from this menu
- Also export this assigned salary details

# Employee management system

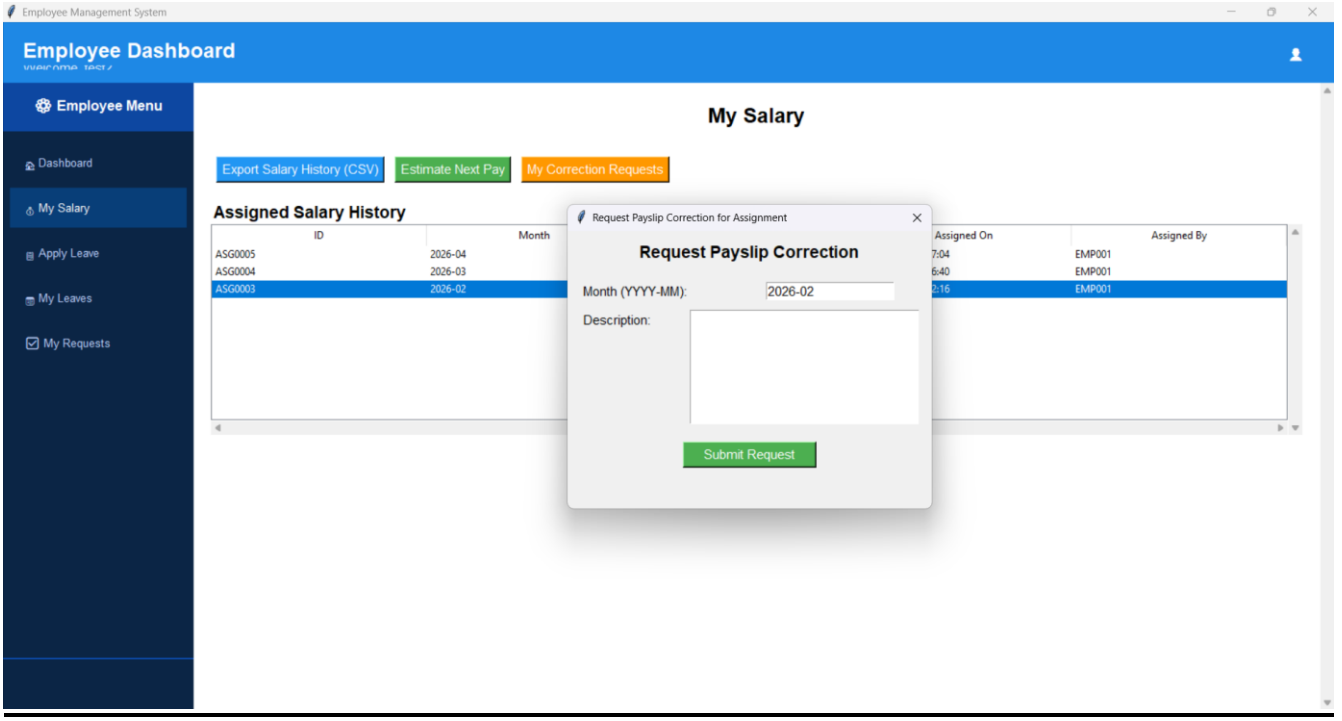
## ➢ Calculate estimate pay module




- Here employee can calculate their estimate salaries

# Employee management system

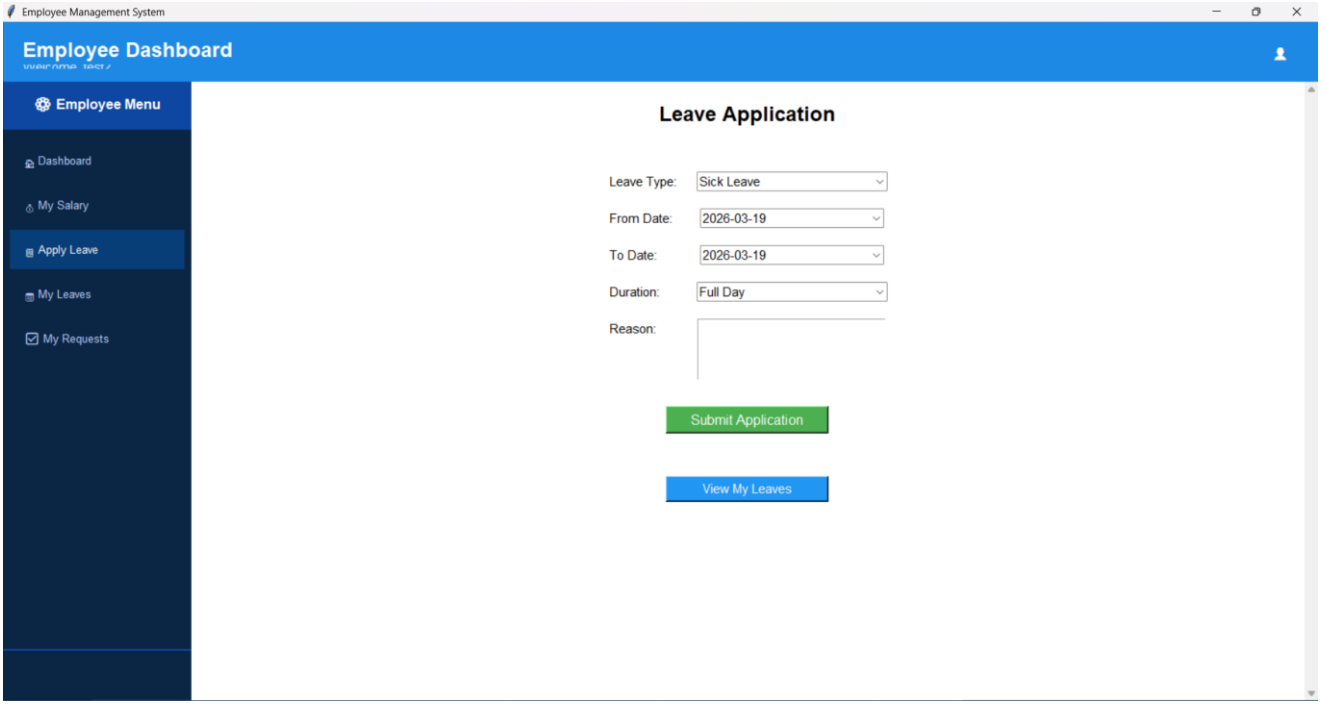
## ➢ Correction Request form




- Using these modal employees can request for correction in their salary

# Employee management system

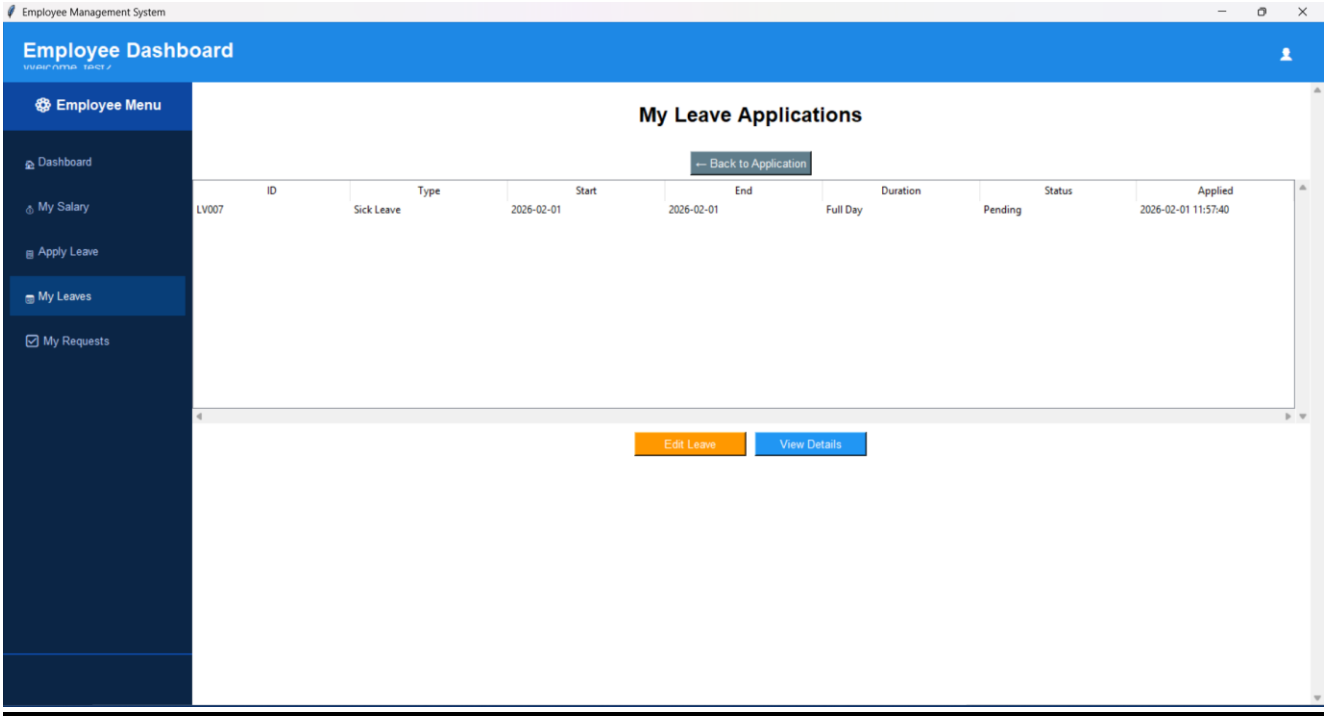
## ➢ Leave application module




- Here employee can apply for leave and using view leave view their leave applications

# Employee management system

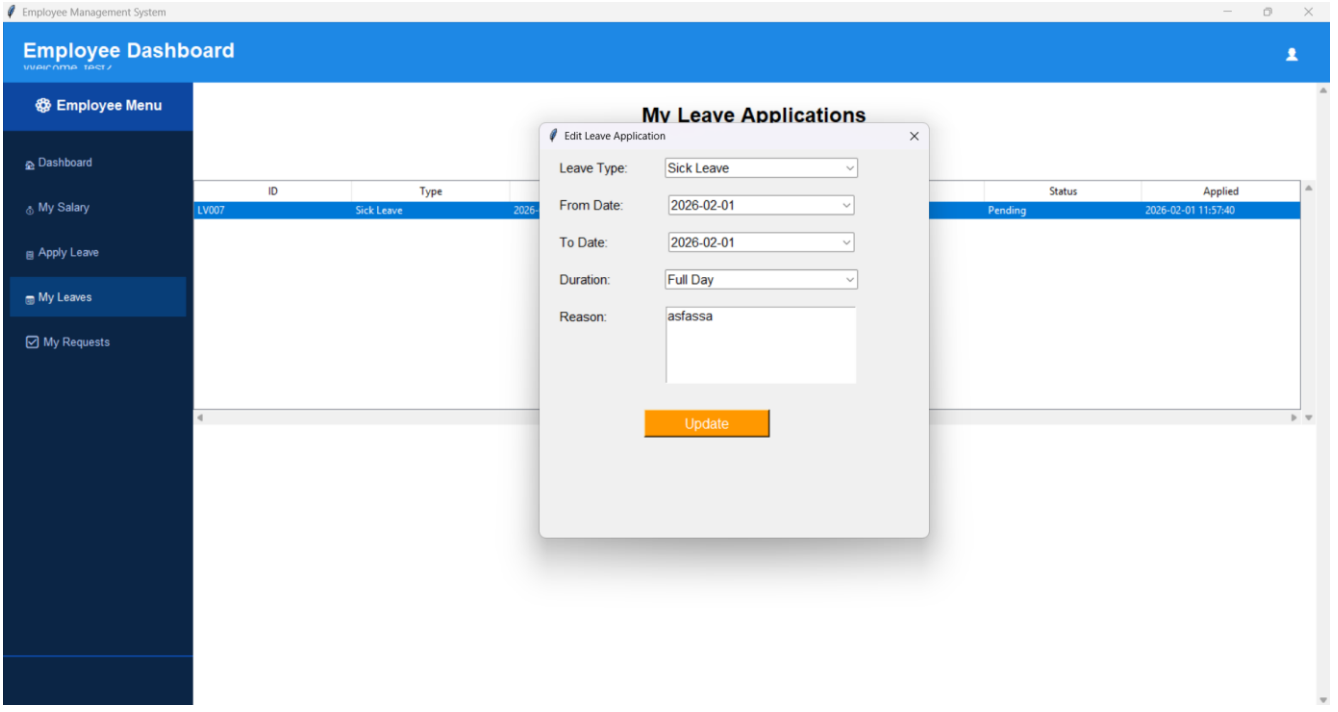
## ➢ Employee leave applications module




- Here also employees can see their leave applications
- Also, employee can edit their leave applications if leave application status is pending

# Employee management system

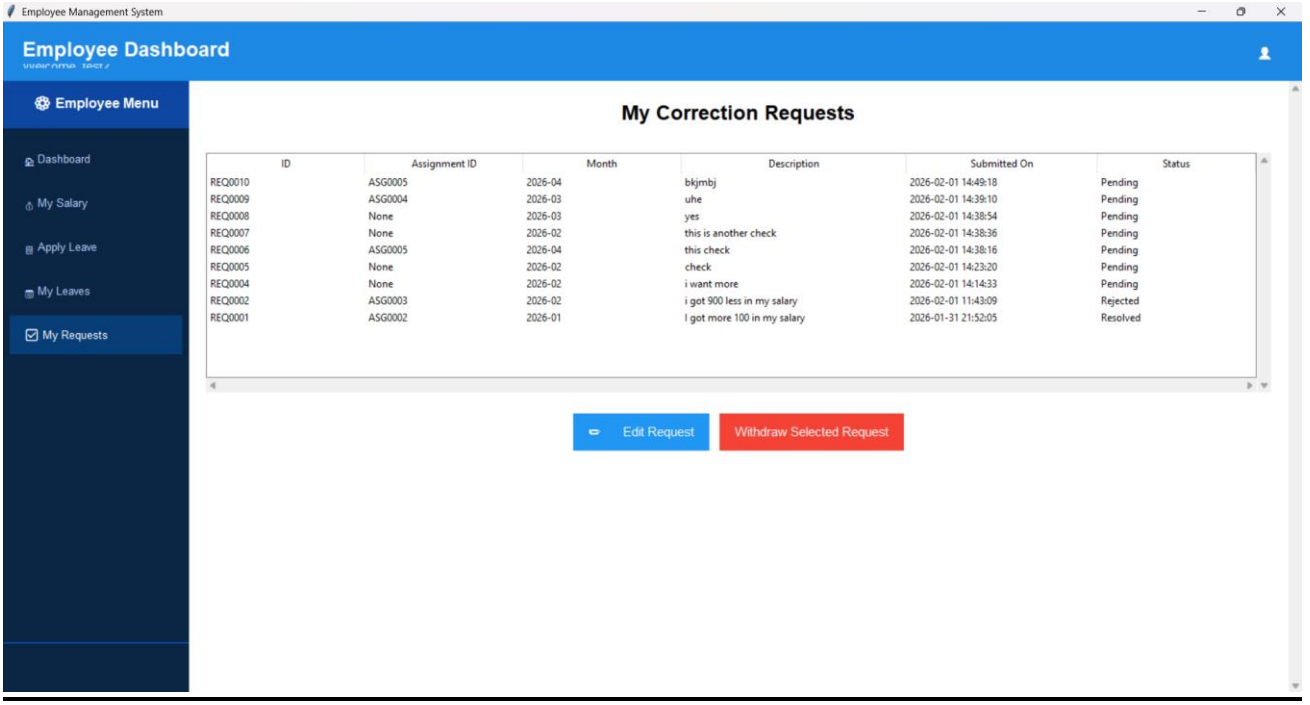
## ➢ Edit leave module




- Using this form modal employees can edit their leave applications

# Employee management system

## ➢ Employee correction list module

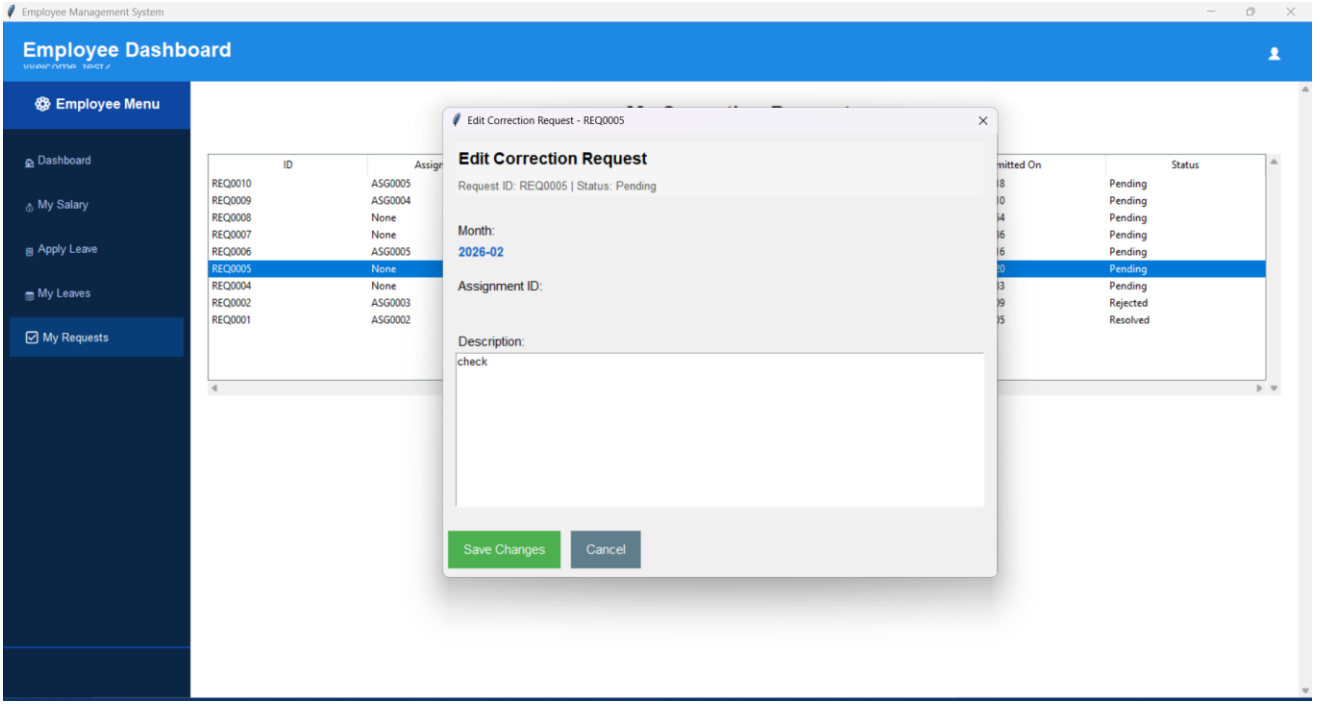



| ID | Assignment ID | Month | Description | Submitted On | Status |
|---|---|---|---|---|---|
| REQ0010 | ASG0005 | 2026-04 | bkjmbj | 2026-02-01 14:49:18 | Pending |
| REQ0009 | ASG0004 | 2026-03 | uhe | 2026-02-01 14:39:10 | Pending |
| REQ0008 | None | 2026-03 | yes | 2026-02-01 14:38:54 | Pending |
| REQ0007 | None | 2026-02 | this is another check | 2026-02-01 14:38:36 | Pending |
| REQ0006 | ASG0005 | 2026-04 | this check | 2026-02-01 14:38:16 | Pending |
| REQ0005 | None | 2026-02 | check | 2026-02-01 14:23:20 | Pending |
| REQ0004 | None | 2026-02 | i want more | 2026-02-01 14:14:33 | Pending |
| REQ0002 | ASG0003 | 2026-02 | i got 900 less in my salary | 2026-02-01 11:43:09 | Rejected |
| REQ0001 | ASG0002 | 2026-01 | I got more 100 in my salary | 2026-01-31 21:52:05 | Resolved |

- Here employees can see salary correction requests and their statuses
- Also, employees can edit correction request if their application status is pending
- Employees can withdraw their correction requests.

# Employee management system

## ➢ Edit correction requests module




- Using this form modal employees can edit their correction requests.

# REFERENCE AND BIBLIOGRAPHY


- **Websites: -**

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-http://www.mysql.com

- **Search engines: -**

-Google search engines
-Mozilla firefox
-Google chrome

- **Reference engines: -**

-www.google.com

-www.w3schools.com